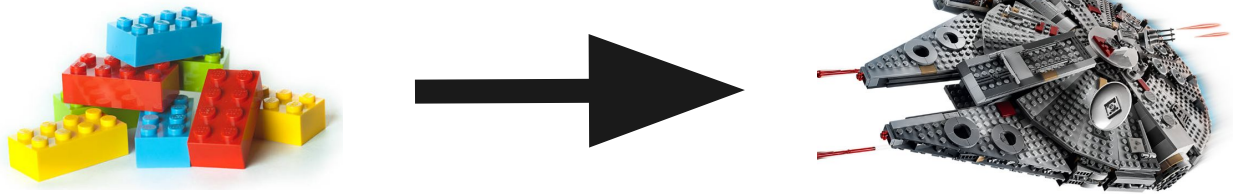
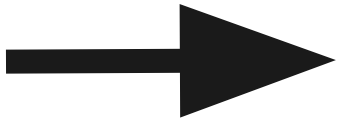
Drone Anatomy: What Are The Lego Blocks Of The Drone?



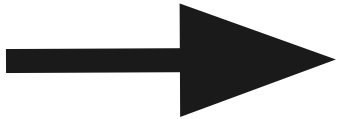
Drone Anatomy: What Are The Lego Blocks Of The Drone?















Drone Motors

The drone motors are what provide thrust to lift the vehicle into the air.

There are two main types of motors used on drones

- Brushed Motors
- Brushless Motors





Drone Motors

Brushed Motors:

Only have two leads, a positive and a negative.

Pros:

- Very simple, just need to power with a voltage and motor will spin

Cons:

- Are not very durable and will wear down much faster than Brushless Motors



Drone Motors

Brushless Motors

Have three leads instead of two.

Pros:

- Very durable, will last much longer than Brushed motors

Cons:

- More complex to run. They need to be driven by a specialized circuit and cannot plug directly into a battery



Drone Motors

Brushless Motors

The specialized circuit that will run these motors is called the ESC, short for Electronic Speed Controller. (We will cover this shortly)

In this camp, we will be using the more durable brushless motors in our drone builds.

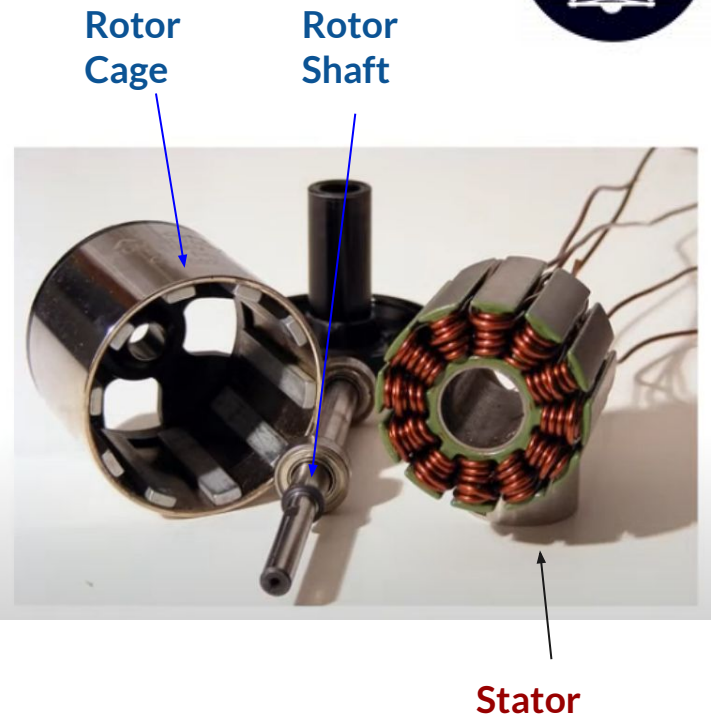


Drone Motors

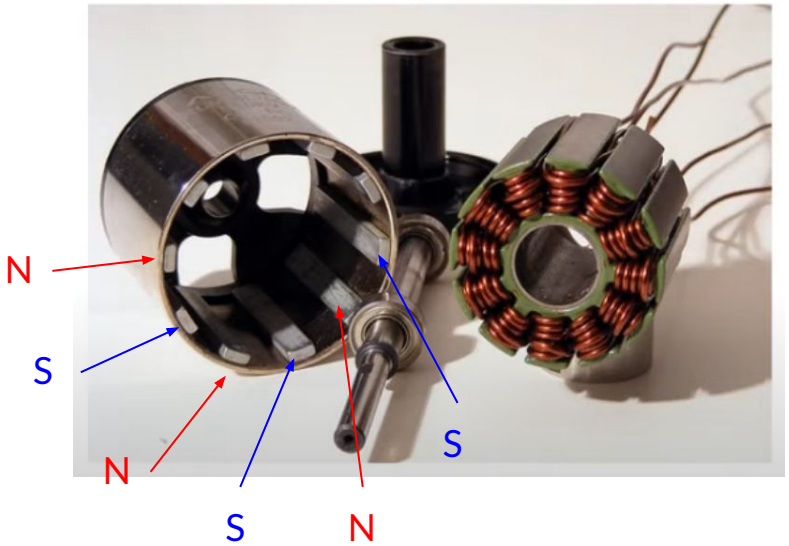
Brushless Motors

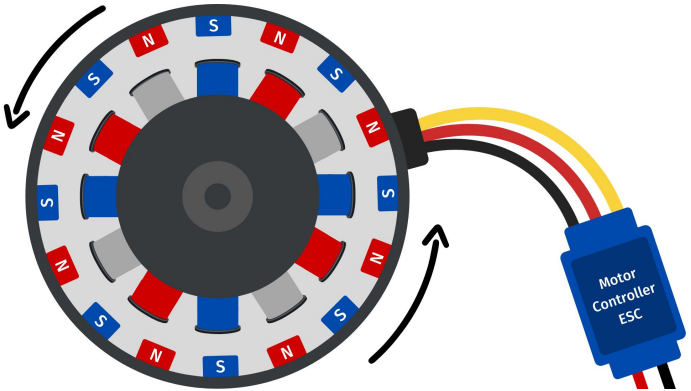
Stator: The part on the right. The stator does not move. Think stator stays). Current runs through the wires to produce magnetic fields that spin the rotor

Rotor: The part on the left is the rotor. This is the part that moves. It has permanent magnets attached to it that try to align with the induced magnetic fields of the stator.

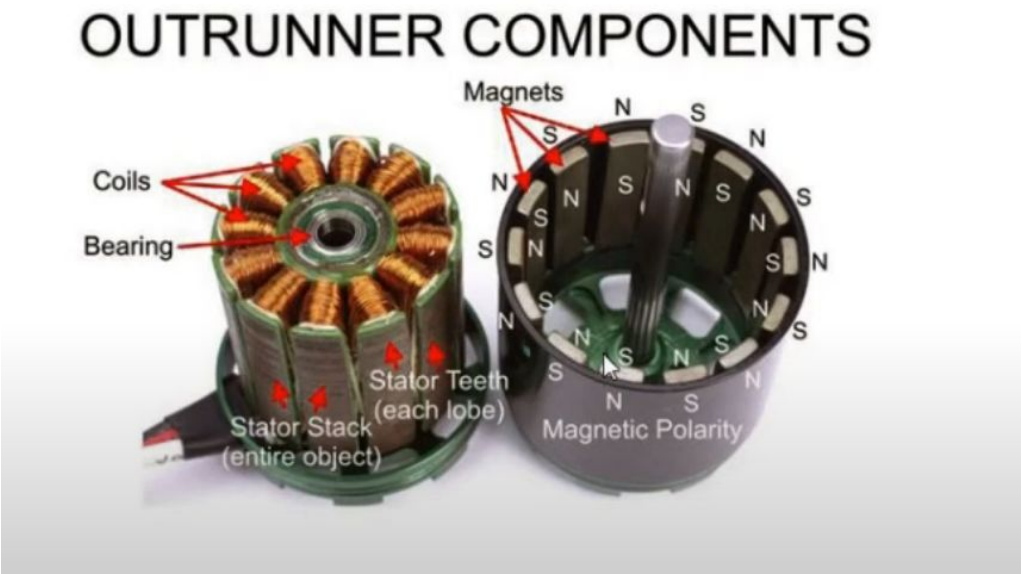


Drone Motors: Brushless





Drone Motors: Brushless



Drone Motors: Brushless





KV Rating

NOT TO BE CONFUSED WITH Kilo-Volts

- KV Rating tells you how fast the shaft will spin per volt with no load (No prop installed)
- 3100KV is the same as 3100 rpm/v
- So if we supplied 1 volt to this motor, we would get it to spin 3100 times every minute
- Higher KV means the motor will spin faster
- Lower KV means the motor will spin slower but have more torque.







LIPO Battery: Many Different Shapes/Sizes



LIPO Battery Overview:

- Lipo stands for “Lithium-Polymer”
- It is essentially the same thing as a Lithium-ion battery
- Battery of choice for RC/Robotics
- Have very high specific energy. (Energy/Mass)
- FLAMMABLE!!!
- Full charge voltage of lipo cell: 4.2 volts
- Rest voltage of depleted lipo cell: 3.7 volts
- Connects to the drone’s Power Distribution Board for providing power to the drone.



LIPO Explosion

The following video is an example why it is extremely important to handle these batteries with extreme caution.

Always store LIPO batteries in a Fireproof Bag!

DO NOT PERFORM THE ACTION SHOWN IN THE VIDEO!



LIPO Battery Overview:

- Full charge voltage of lipo cell: 4.2 volts
- Rest voltage of depleted lipo cell: 3.7 volts
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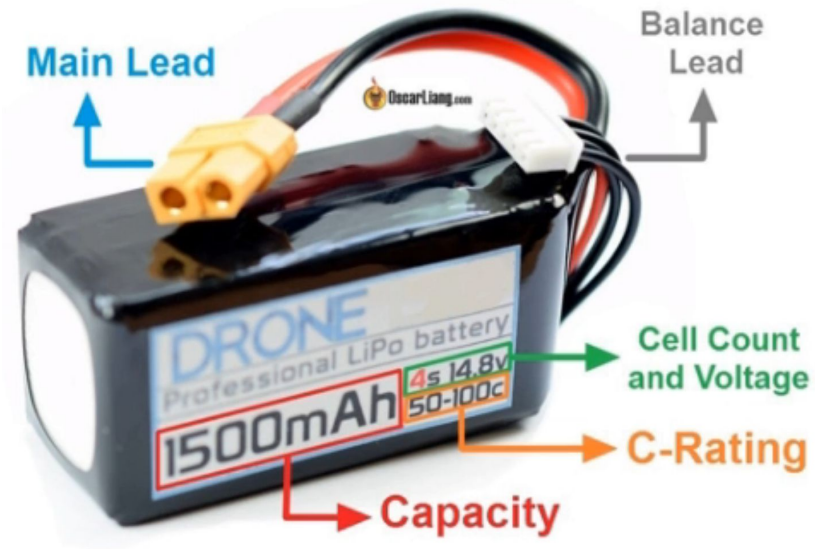



OVERLANDER
3350 mah
11.1V



Power Distribution Board (PDB)

Units of the Battery



Capacity

Capacity is a measure of the amount of electric charge (Coulombs) that is stored within the battery.

Typical units: mAh or Ah

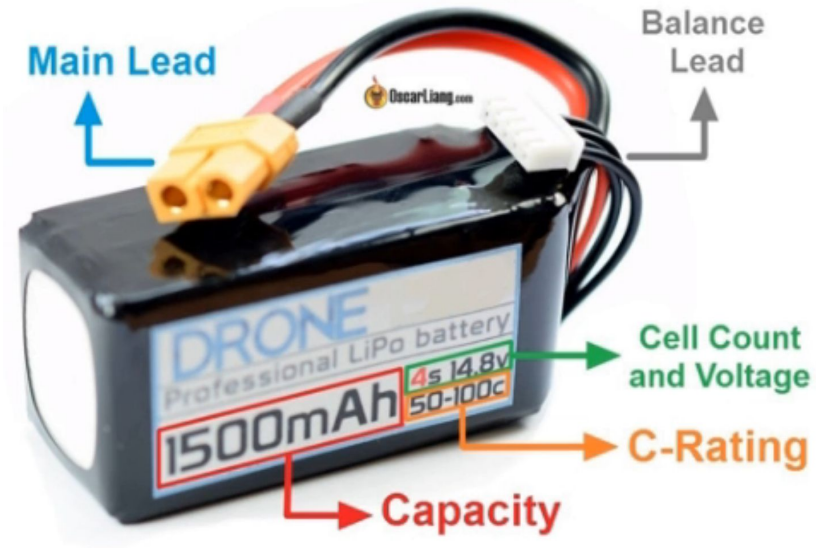
A=Amp=Coulomb/second=C/s

mAh= milli-Amp hour

mAh=(1/1000)(C/s)(1 hour)(3600s/1 hour)

mAh= 3.6 Coulombs

So mAh or Ah is simply the amount of Coulombs that are in a battery!



C-Rating

The C-Rating vies us knowledge of the maximum current draw that the battery can safely endure.

Units of C are inverse hours (1/hour)

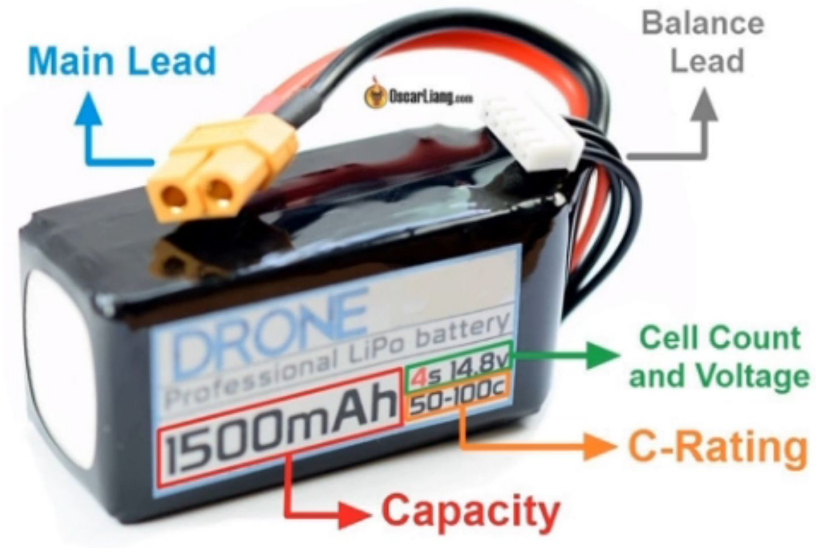
Example with this battery (50C):

1500mAh=1.5Ah

Maximum Current Draw=(1.5Ah)(50C)

Or (1.5Ah)(50/h)=75A

So this battery can safely have 75 amps continuously drawn from it!



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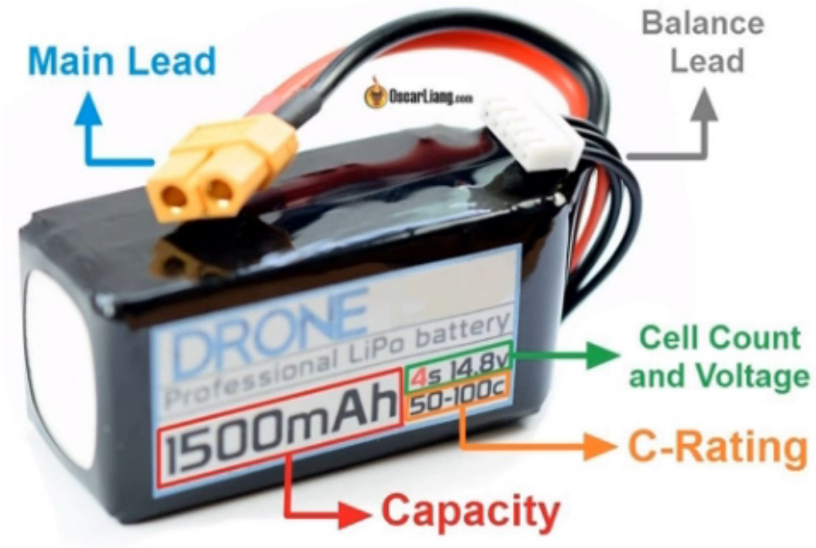
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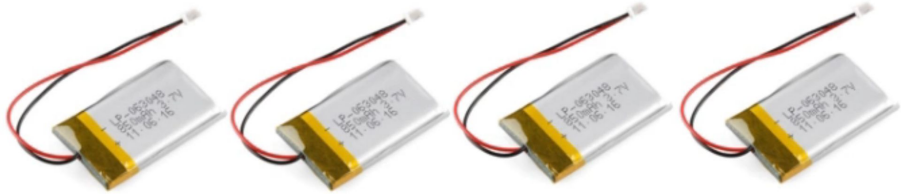
Or (1.5Ah)(50/h)=75A

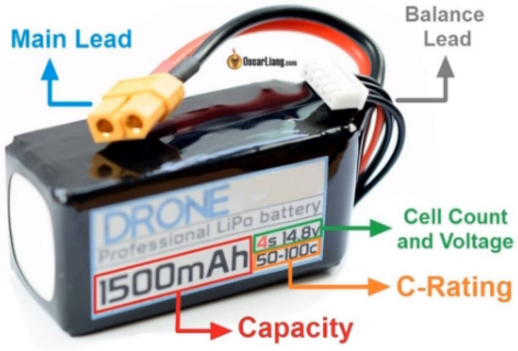
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Lipo Cells

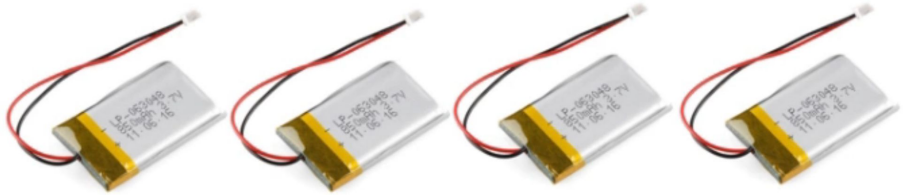
- Lipo battery packs are composed of multiple individual lipo cells
- Full charge voltage of lipo cell: 4.2V
- Nomical voltage (depleted): 3.7V
- S and P on battery packs: S means “Series” and P means “Parallel”
- The number before S or P indicates number of cells
- Voltages of lipo cells add if in series
- **This battery is 4S, so it is made of 4 lipo cells connected in series**

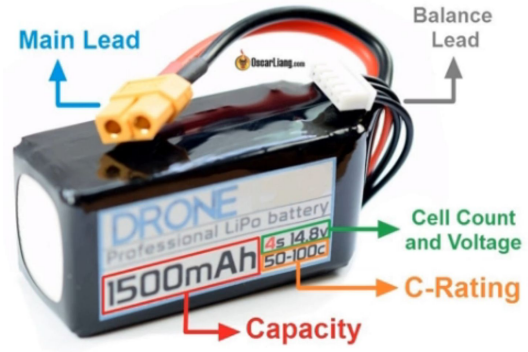




Lipo Cells

Full Charge Voltage Of This Battery? Any Guesses?





Lipo Cells

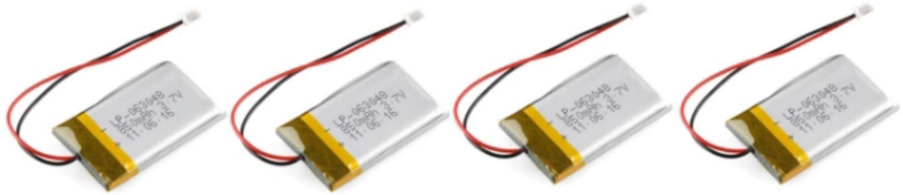
Remember:

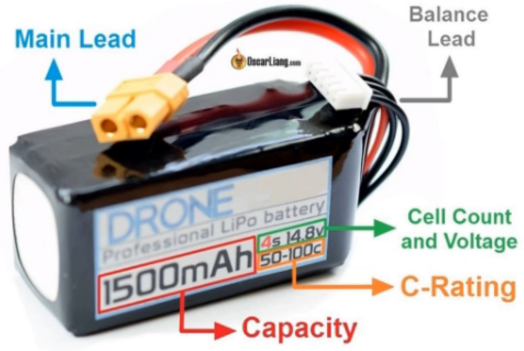
Full Charge=(number of cells in series)*(full charge voltage)

Full Charge=(4)(4.2)=16.8 Volts

Depleted Charge=(number of cells in series)*(depleted charge voltage)

Depleted Charge=(4)(3.7)=14.8 Volts





Lipo Cells

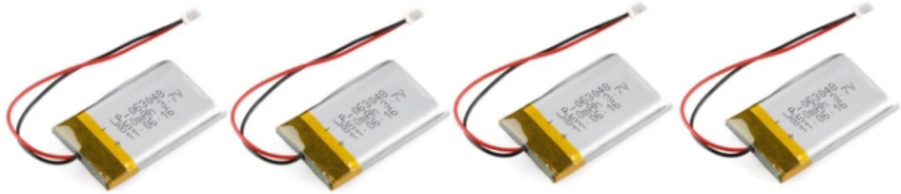
The large red and white wires provide the full voltage of the battery pack. This is what plugs into the power distribution board.

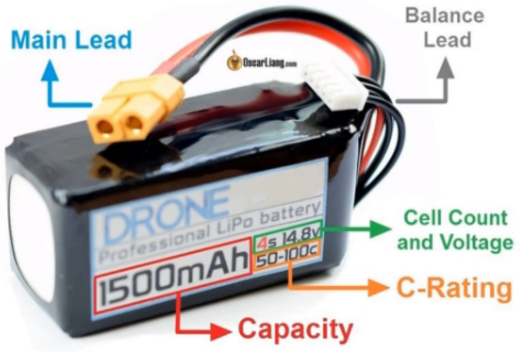
You also have access to each individual cell through the smaller black wires in this battery example.

You need access to the individual cells so that you can make sure each cell voltage is at the same level.

If the cells aren't the same voltage, there will be a voltage difference, which will cause current to run in between individual cells, instead of the desired state of current running through the entire battery pack.

Your battery charger will balance each individual cell to the same voltage so this doesn't happen.





Flight Controllers (FC)

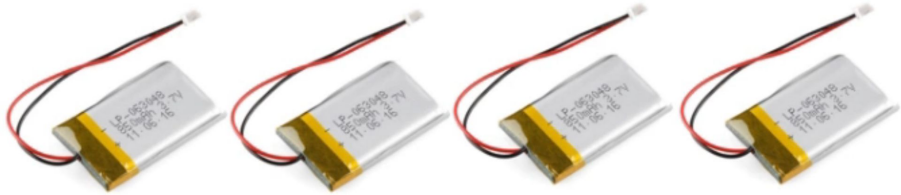
The large red and white wires provide the full voltage of the battery pack. This is what plugs into the power distribution board.

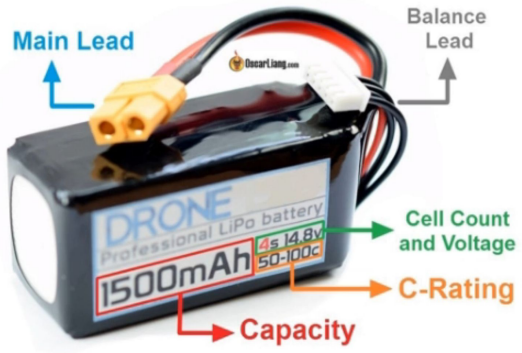
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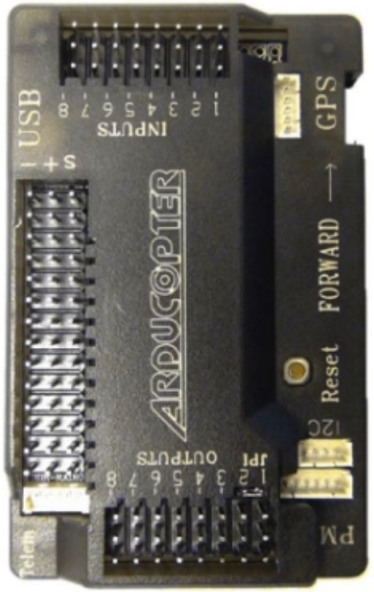




Flight Controllers (FC)







Flight Controllers (FC)

- FCs are the control center of the drone
- They are the central hub that ALL electronics connect to (motors/ESCs, GPS, Telemetry, RC etc)
- This is where the drone's firmware runs (ArduPilot)
- The Firmware is the code that interacts with and commands the hardware



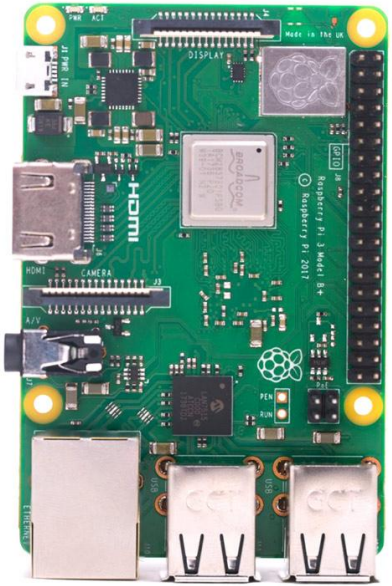

pixhawk


FLYSKY




pixhawk

Companion Computer (CC)



Companion Computer (CC)



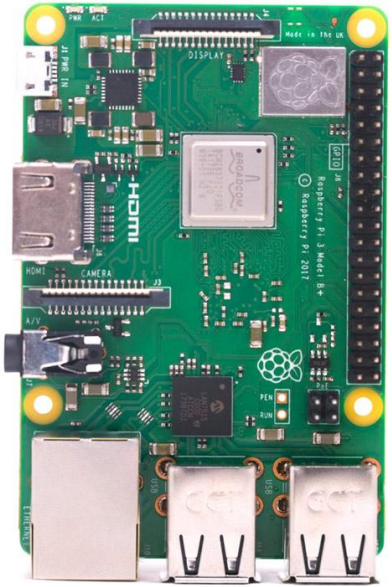
Companion Computer (CC)

How many lines of code do you think were required to make this autonomous taco drone delivery mission… any guesses???



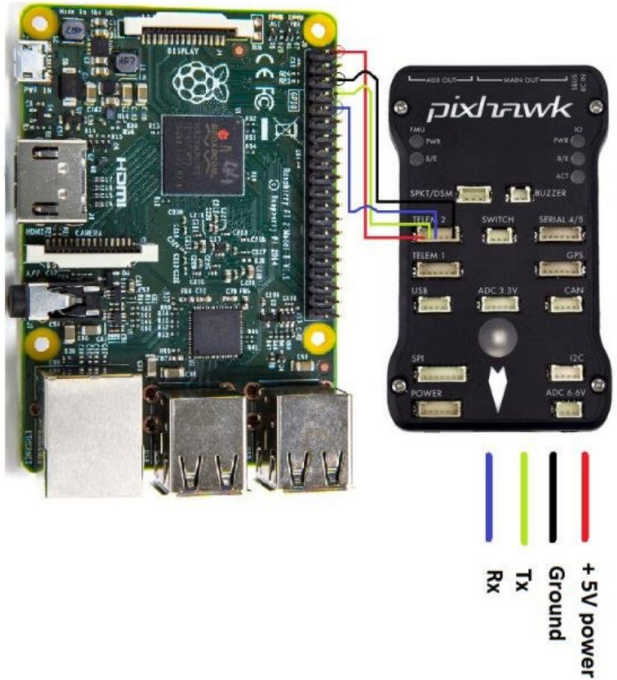
Companion Computer (CC)

- Companion computers unlock so much possibilities. You're only limited by your imagination. Here are some examples:
- 4G drone for limitless range (control drone from 1,000 miles away)
- Artificial Intelligence
- Object Recognition
- Simple Python Scripting to control drone
- So much more!
- These are things that a flight controller alone cannot perform.

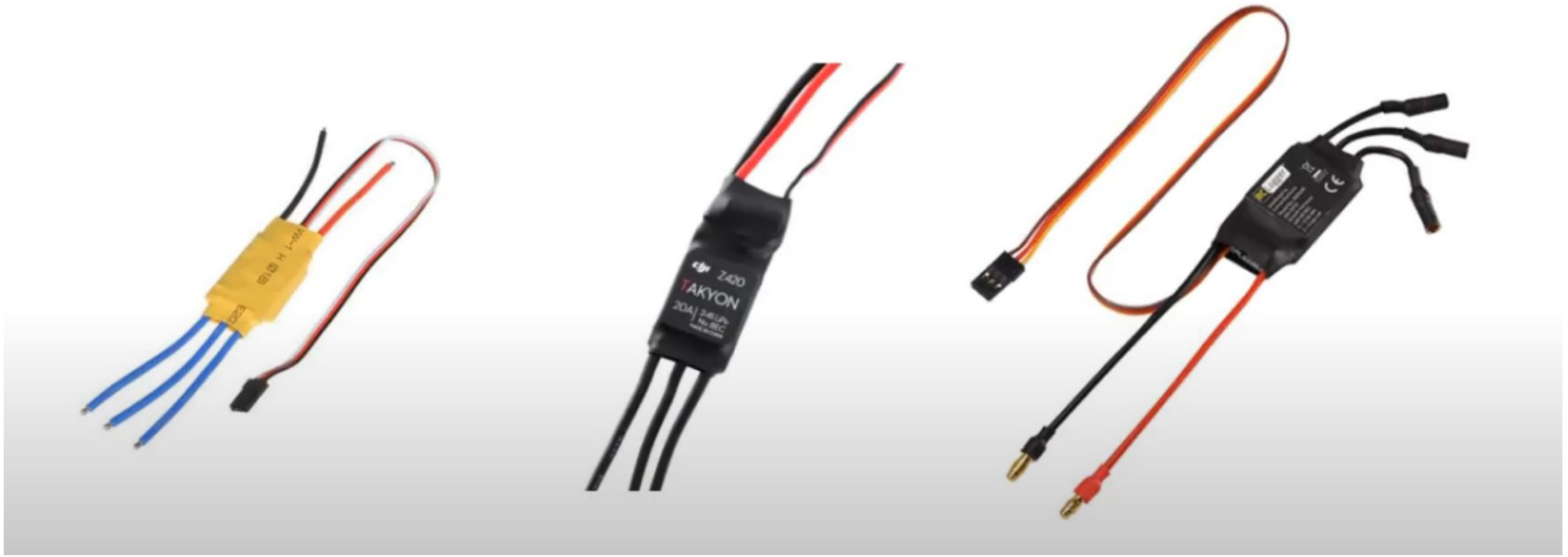


Companion Computer (CC)

- The companion computer also plugs into the flight controller just like the rest of the drone hardware.
- Typically it is via a UART connection
- Your drone kit has a special connector to make this UART connection.
- Later in the camp, we will explore controlling the drone from the raspberry pi for autonomous flight.



ESCs



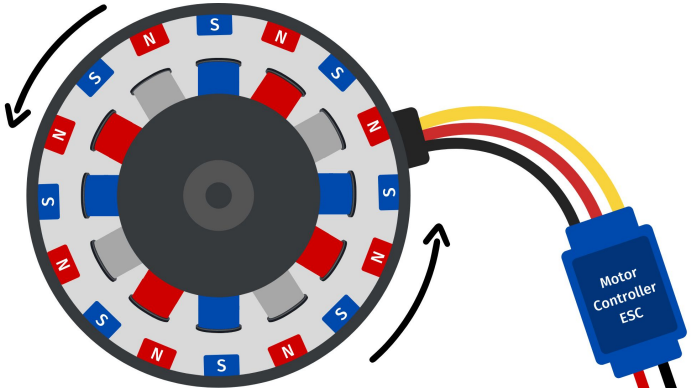
ESCs

- Red and Black wires connect to the Power Distribution Board
- Signal wire connects to the flight controller (FC)
- Flight controller will output a PWM signal to the ESC
- ESC then electronically digests the PWM sent from the FC output and produces an ESC output that will drive the motor's stator and spin the props
- Three output wires connected to the motor



ESCs change the magnetic field on the stator of the motor





ESCs



ESCs



Special Units/Info on ESCs



Voltage Rating

- ESCs will tell you what types of LiPos you can use with them
- Here, we see 2-4S. So we could use 2, 3, or 4 cell LiPos (from 7.4V to 14.8V) with this ESC safely.



Current Rating

ESCs can only handle so much current running through them

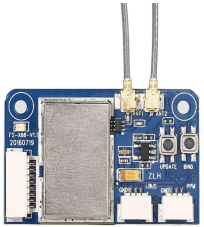
This ESC shows 30A, so when designing our drone build and spec'ing out parts, we'd need to make sure that our motor would never draw more than 30A

It's typically a good idea to select an ESC that is 20% higher current rating than the maximum expected current draw of the motor



RC and Telemetry



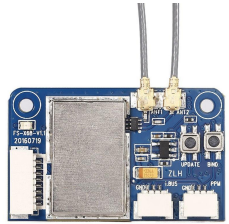




RC & Telemetry Overview

- Both RC and Telemetry are used to communicate information wirelessly with our drone
- Telemetry communication is bi-directional, RC communication is unidirectional (from RC controller to receiver)
- RC typically uses the 2.4 GHz band
- Telemetry typically uses 915 MHz band in the United States and 433 MHz band in Europe
- RC and Telemetry involve to pieces of hardware that are paired. One piece is drone-side, and the other ground side (RC controller is off drone, and RC receiver is on drone)

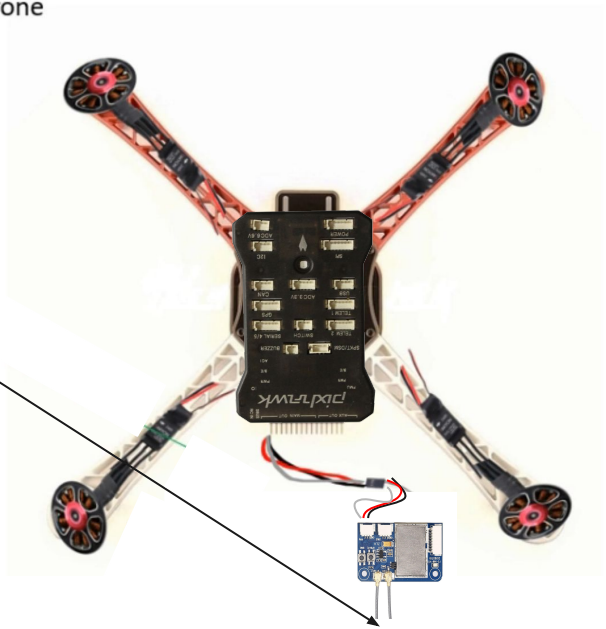


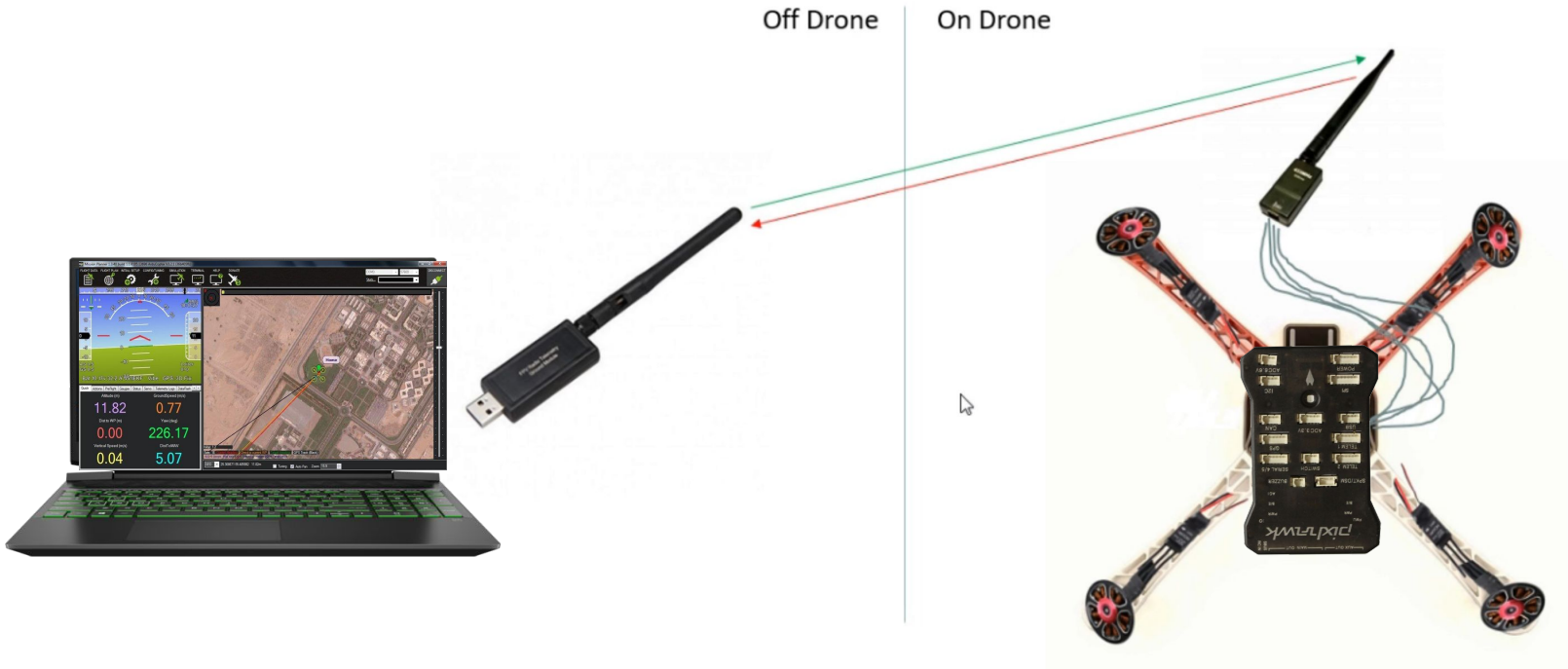


Off Drone

On Drone



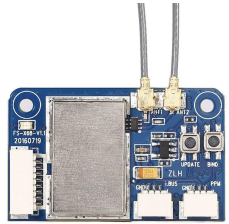


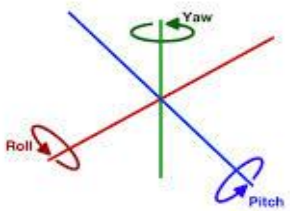

Off Drone
On Drone
11.82
0.77
0.00
226.17
0.04
5.07

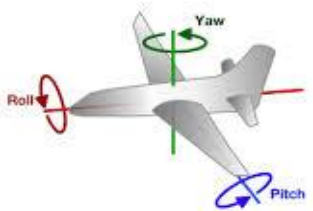
RC

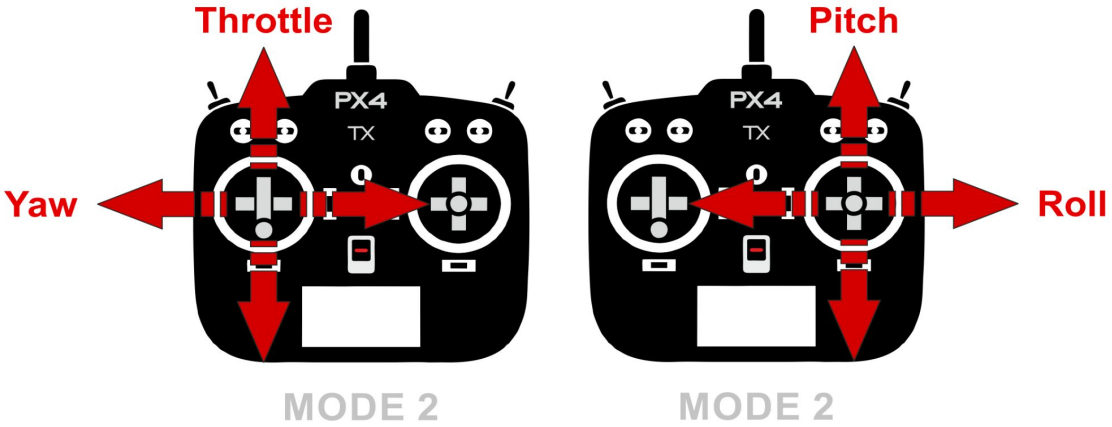
- Used to control the drone when in a manual flight mode (that requires a pilot. There are autonomous flight modes that do NOT require an RC controller to fly drones).
- Price increases exponentially with the amount of channels your RC setup has.
- Minimum amount of channels for most drone applications: 6 channels
- 4 channels are dedicated to Roll/Pitch/Yaw/Throttle
- Remaining channels can be used to control various things on the drone
- Channel 5 typically dedicated to switch flight modes of drone






Yaw
Roll
Pitch

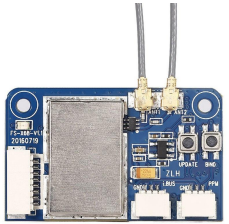

Yaw
Roll
Pitch


Throttle
PX4
TX
Yaw
MODE 2
Pitch
PX4
TX
Roll
MODE 2

Knob

Switch





Telemetry

- Used by ground control stations to communicate with airborne drones (Ground control stations are basically off-drone computers that have a communication link with the drone via telemetry)
- Can gather information from drones (like location, speed, battery voltage, system health, flight mode etc)
- Can set variables (Example: setting flight modes)
- Can control the drone (Example: Move to this GPS waypoint on the map)



Props



Overview

- Selecting propellers (props) should be done with motor and battery selection in mind.
- Different prop specs can produce dramatically different current draws and thrust.



Props: Diameter

- Specs of the props will be provided in 4 digits of numbers. For example: 1045
- The first two numbers are for the diameter of the prop in inches. So a 1045 prop is 10 inches long
- So what about the 45 part?





Props: Pitch



- The last two numbers are for the pitch of a prop
- Imagine a screw turning into wood. The more aggressive the pitch, the farther the screw will turn into the wood in one turn.
- That is the same principle being measured with prop pitch. So on our 1045 prop example, 45 indicates the prop would turn 4.5 inches in one complete turn if it were able to turn into a solid medium
- The higher the pitch, the more torque you will generate (the more air it will push when spinning)





Props: Pitch



- Different types of props can drastically affect the current draw of your setup.
- Using the same LIPO battery and motors, using a 1045 prop will draw much more current than 8045 props. 1045 props will also produce more lift.
- Be careful not to over spec the props you are using (AKA, be careful not to choose too large of a prop.) Make sure your motors/ESCs can handle the amount of current the motors will draw with the props being used
- Most motors will tell you suggested prop dimensions to use with their model.





Frames

- There are many different types of frames/drone types
- Frames can alter in motor counts, and different arm configurations for the same number of motors
- Generally, having more motors means more lift (able to have heavier payloads)
- Multirotor=multiple motors. Tri-copter (3 motors), quadcopter (4), hexacopter (6), octocopter (8) etc





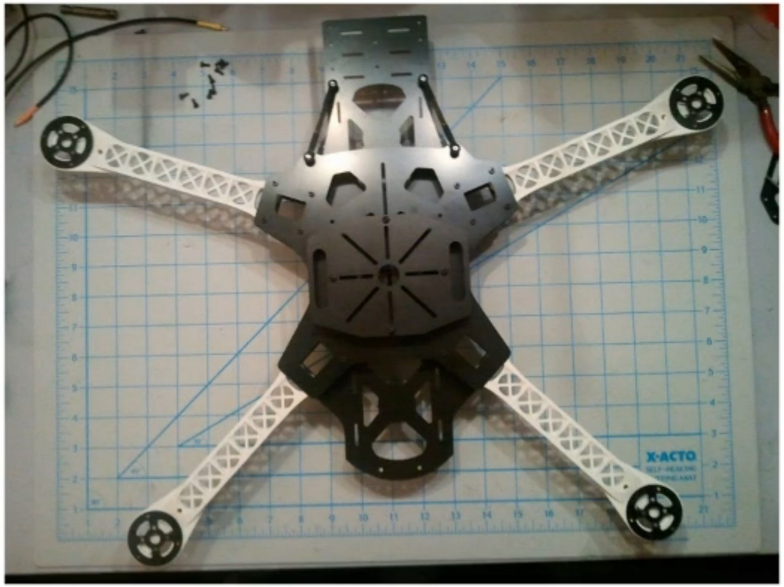


Different Types of Quad-Copters

- Quad=4
- Quadcopters have 4 motors
- There are 3 main types of quad-copter frames



Deadcat Frame



PROS

Have a lot of mounting points on the frame, making it easy to place all the necessary components

CONS

Not optimized for aerodynamics, and are for that reason little less efficient. Also weigh more than their quadframe counterparts, so won't fly as long as other models.

H Frame

PROS

Simple configuration

Easy to make from scratch

CONS

Mostly lighter than dead-cat frames, but heavier than X Frames



X Frame

PROS

Very aerodynamic

Lightest of the quadframes

CONS

Less space for mounting drone components (like GPS, flight controller, battery etc)



Tri-Copter

Not very popular frame of multi-rotor, but do exist

Lightweight

No redundancy. If one motor fails, the tri-copter is crashing



Hexacopters and Octocopters

Hexa=6 motors

Octo=8 motors

The more motors a multi-rotor has, the more lift it will generate. The more lift it can generate, the heavier the potential payload.

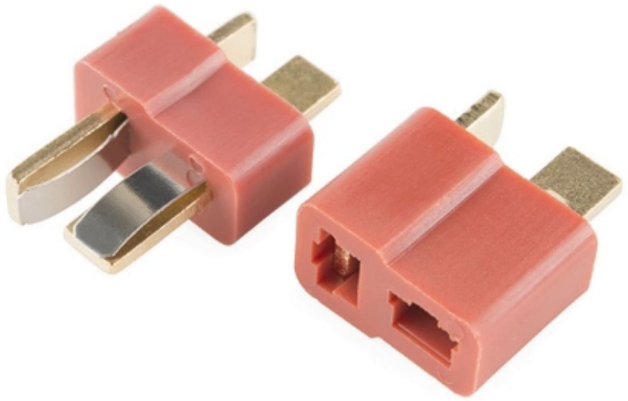
-> Drone applications requiring heavy payloads should use higher motor multirotors (example applications: drone delivery, or pesticide crop spraying drones)





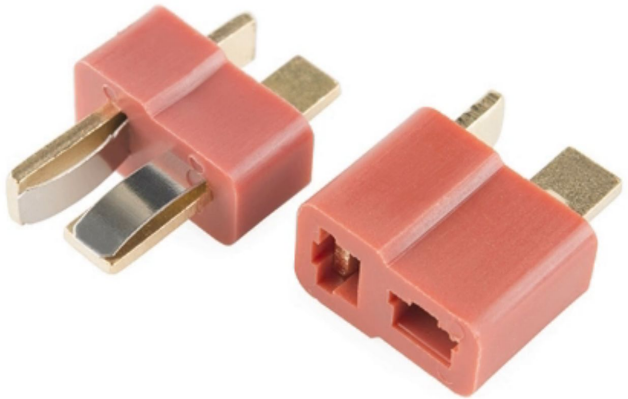
Battery Connectors

- Many types of battery connectors
- All consist of 1 female end and one male end
- Connectors have maximum current ratings
- Generally don't have to worry about the current rating of the connector, since the lipo batteries come pre-soldered with the correct connector on it.



Dean's Connector (T-plug)

- Maximum current draw: ~60A
- Can vary depending on the authenticity of the connector
- Female end typical on LIPO battery side.





XT-60

Maximum continuous current rating: 60A

Male end typically on the LIPO battery side

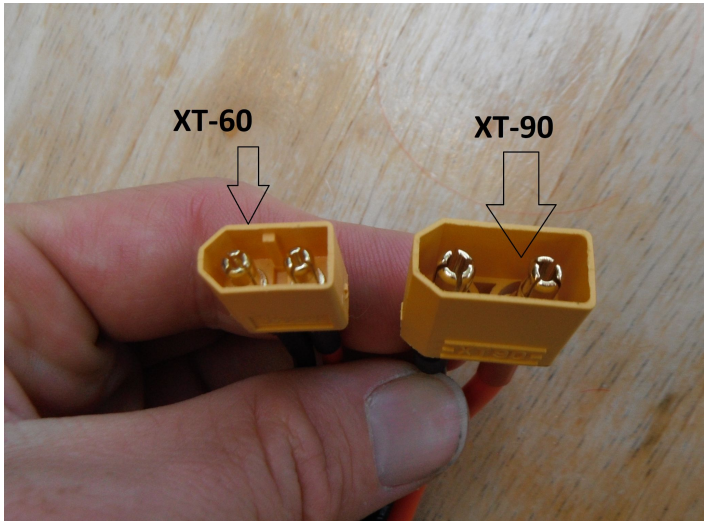




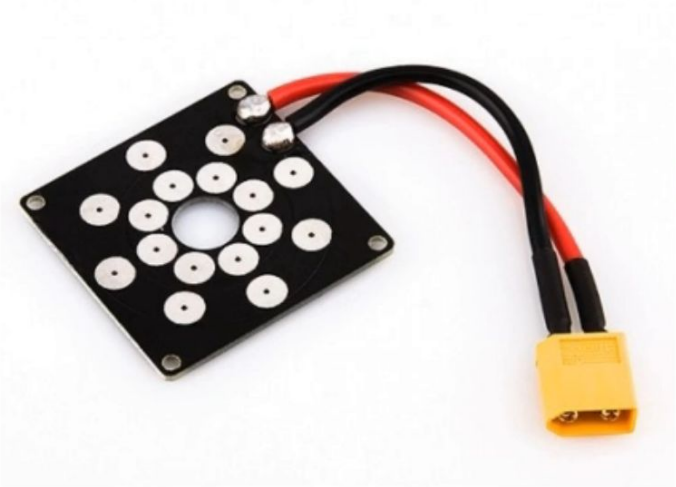
XT-90

Maximum continuous current rating: 90A

Male end typically on the LIPO battery side



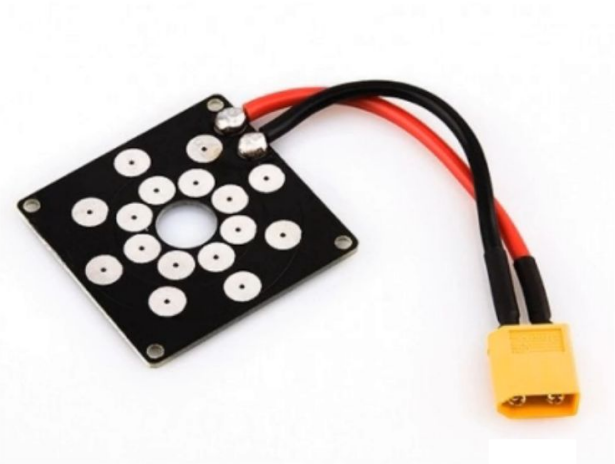
What if PDB has a different connector?







What if PDB has a different connector?









GPS



GPS

For Manual operation of the drone, GPS isn't required, since the drone simply goes where the pilot is commanding it to go with the RC controller. **But what about for autonomous missions without a pilot?**

For autonomous drone missions, we need a way for the drone to determine its position in 3D space.

That is the function of the GPS. We can define waypoints for the drone to fly to, and with GPS can autonomously fly to the waypoints. *NO PILOT REQUIRED!*

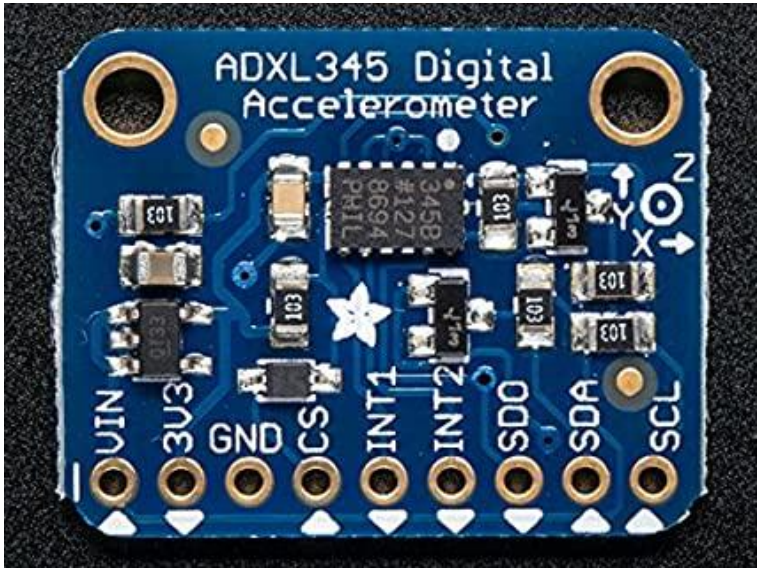


Basic Required Sensors for Flight

To fly a drone like a quadcopter, you really only need:

1. Accelerometer (senses the tilt of the drone)
2. Gyroscope (senses rate of angular speed)

But this is only practical for manual flight. For autonomous missions, we need to combine these sensors with more robust methods and sensors for determining the drones position in 3D space



GPS

Used in conjunction with a magnetometer (to determine YAW orientation)

With GPS, drone knows where it is in 3D space

To travel to another location, simply travel from known current position to target waypoint



PROS:

- Self adjusts its errors (If a gust of wind blows the drone off course, a new reading will simply show where the new location of the drone is in 3D space
- Cheap

CONS:

- Needs to be outside to access satellites
- Can produce faulty readings from large structures (like if too close to a building)